# **L1a**: Course Orientation, Markets, Assets, and the Decision Problem

CHEME 5660 Cornell University

# Disclaimer and Risks

**Training only.** This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

**Risk.** Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

**Responsibility.** You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

# Objectives for Today

Today we place the course in context, show that you are already a participant in the markets we'll study, and we'll close with a question for next time.

## Today's concepts

- Course roadmap: from comparing payments over time to making and testing financial decisions under uncertainty.
- Why this concerns you: most U.S. adults hold a retirement account, and the markets those accounts invest in move several hundred times more cash per day than industries that employ many engineers.
- The decision problem: investment choices pay at different times, so we need a rule for comparing them.

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**Note:** this lecture has no notebook and no notes packet. These slides are the record.

# How This Course Runs

Across twenty-eight lectures, we move from the financial assets themselves to how we value, combine, and trade them. Six problem sets ask you to apply these ideas to real data and practical decisions.

- Materials: each week is posted as a zip file on the repository releases page, or you can clone the repository. Released weeks are frozen: corrections are announced, never made silently.
- Tools: Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. Set these up once, at the start of the semester.
- Questions: ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

The course ends with your automated trader using live data from Alpaca Markets.

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**Policies:** grading, attendance, academic integrity, and accommodations are posted in full

# How You Will Work

Throughout the course, you will repeat the same cycle: ask a question, use a model to explore it, and test the answer against evidence.

- Each idea answers a question. New concepts are introduced through the financial problems they help us solve.
- Code is our laboratory. Most of the code is provided; you will change assumptions, inspect the results, and explain what the model is doing.
- Evidence decides. A model can be mathematically correct and still fail. You will compare its claims with data and revise it when the evidence disagrees.

You will learn to explain what a model predicts, test that prediction against data, and recognize when the model fails.

# Most of You Will Eventually Own a Portfolio

Most of you may not have a retirement account today. But roughly 61% of U.S. adults have a tax-preferred account, such as a 401(k) or IRA. That makes investing widespread, not an activity limited to wealthy people. (2025)

- A retirement account holds a portfolio: a mix of investments.
- Many account holders keep the default mix selected for them.

Keeping the default mix is still an investment decision.

**Source:** Federal Reserve SHED, 2025, table 28

# The Federal Open Market Committee

A committee of twelve people meets eight times a year, and its decisions change the value of everything in that account.

- Thursday: what the committee controls, what it does not, and why that distinction determines which rate you use to find what an investment is worth.
- September 15 and 16: the committee meets. We will be three weeks into the course, and we return to the decision in week four with the tools to describe its effect.

Sixty seconds on this today. The details are Thursday's lecture.

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**Link:** FOMC calendar and meeting materials

# Market Scale Compared to Two Industries

Two industries this college trains people to enter, compared with the U.S. financial markets. The annual industry figures are divided by 252 trading days.

| | Per year | Cash per trading day |
|---|---|---|
| **Global biopharma** | USD 666B | USD 2.64B |
| **Global batteries** | USD 181B | USD 0.72B |
| **Together** | | USD 3.36B |
| US debt | | USD 1,209B |
| US stocks | | USD 608B |
| Prices paid for options | | USD 37B |
| **US markets together** | | USD 1,854B |

These markets move about five hundred times more cash per day. The work is engineering: estimation, uncertainty propagation, optimization, and control.

**Sources:** market sizes 2025 (Fortune Business Insights); turnover 2024–2026 (SIFMA, Cboe)

# Price Paid Versus Value Controlled

The options row is not like the other two. The USD 37B is the price paid for the options. That price, called the premium, is not the total value of the shares those options control.

Each contract is an option on **100 shares**. At 60.4 million contracts a day, the shares underneath are worth about USD 4 trillion:

$$\underbrace{\text{USD } 37\text{B}}_{\text{price paid (premium)}} \Big/ \underbrace{\text{USD } 4{,}000\text{B}}_{\text{value controlled}} = \underbrace{0.92\%}_{\text{about 92 cents per \$100}}$$

If every contract were exercised on the day it traded, that USD 4 trillion of stock would change hands, which is more than twice the debt and stock rows combined. The previous table reports the price paid instead, because that is the cash that actually moves.

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**Careful:** $3.9\text{M} \times 100 \times 7{,}692 = \$3.0\text{T}$ of that is one index, settled in cash. It never delivers a share.

# The Three Kinds of Asset

The investment choices in this course fall into three broad groups. We name them today and examine each one later, when the course needs it.

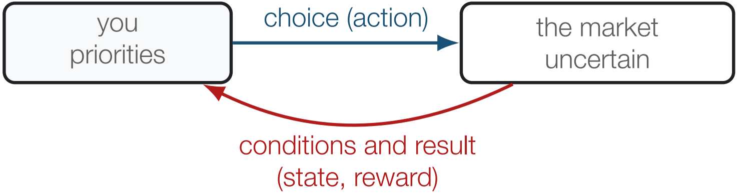
| Kind | What you own | Where the course takes it apart |
|---|---|---|
| Debt | a promise of payments on stated dates | Unit 1, weeks 2 and 3 |
| Equity | ownership of what remains after debts | Unit 2, weeks 3 to 7 |
| Derivative | a contract tied to another asset's value | Units 3 and 4, weeks 8 onward |

Risk and potential reward both increase down the list. Debt promises a fixed schedule but the borrower can fail to pay. Equity promises no fixed payments and is worth whatever remains after the company pays what it owes. A derivative depends on another asset, so its possible outcomes vary most widely.

**Today:** we use only the first kind, and we do not use its name until later.

# The Decision Problem

Every investment decision follows the same pattern: you observe the market, choose what to do, and see what happens. In a decision model, these are called the state, action, and reward.

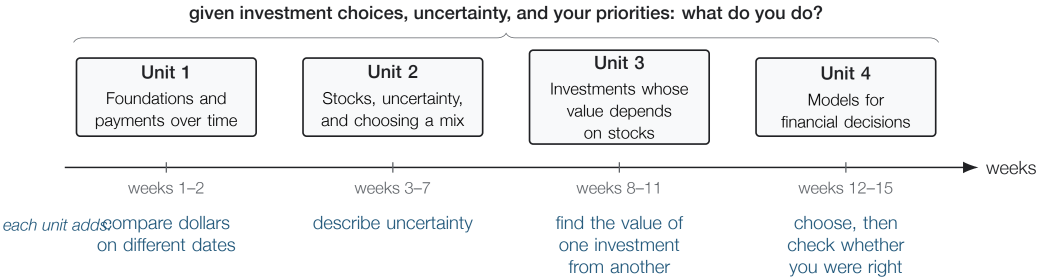



To use this model, we must compare payments over time, describe uncertainty, find values that depend on others, and choose. These are the four units.

**Careful:** each arrow in the diagram is a modeling choice.

# The Four Units of the Course

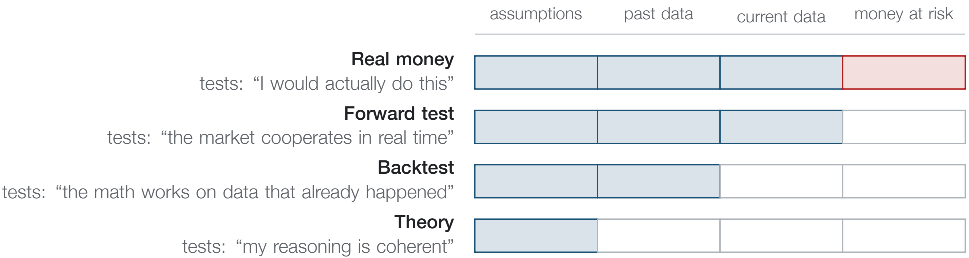
Each unit adds exactly one capability, and the unit that follows it cannot be built without that capability.




Today we start with the leftmost unit, and we stay there through Thursday.

# The Validation Ladder

Four ways to test an idea. Each stage uses evidence the one below did not see, and each can reveal a failure that the earlier stages cannot.




A backtest runs on past data. A forward test runs on actual current data with no money at stake, using current market data from a service such as Alpaca.

Skipping a stage means discovering a failure later, perhaps with real money at stake.

# Five Promises for $10,000

Each of these costs $10,000 today, and each is calculated using an interest rate observed in the market yesterday. The names come later; for now, only the payments.

| | |
|---|---|
| **A** | $10,028 back, four weeks from today |
| **B** | $236 every six months for 10 years, then $10,000 |
| **C** | $264 every six months for **30** years, then $10,000 |
| **D** | $271 every six months for 10 years, then $10,000 |
| **E** | $121 every six months for 10 years, plus an inflation adjustment on every payment and on the $10,000 |

**Which one do you take? Hands up.**

**As of:** published rates for 2026-08-18, each scaled to a $10,000 outlay

# What the Five Promises Are Called

| | What it is called | Annual rate |
|---|---|---|
| A | 4-week Treasury bill | 3.63% |
| B | 10-year Treasury note | 4.71% |
| C | 30-year Treasury bond | 5.28% |
| D | investment-grade corporate bonds | 5.42% |
| E | 10-year inflation-indexed Treasury (TIPS) | 2.41% real |

You compared all five without knowing any of these names, because each name labels a schedule: a list of payments and their dates. Thursday's lecture starts from that definition.

**Careful:** D is an index of many investment-grade corporate bonds, not one bond. E's rate is *real*, not nominal.

# Three Questions We Cannot Answer Yet

Most hands went to **C** or **D**, which is reasonable given the information on the previous slide. That slide left out three things.

- Why not C? It pays the most of the three government rows, but you may need the money before year thirty. If you sell early, you sell at whatever price interest rates at the time allow. Week 2.
- Why not D? It pays more than any other row. Those borrowers pay 0.71 percentage points more than the government over the same ten years ($5.42\% - 4.71\%$), which is what the market charges for the risk of not being paid and for holding something harder to sell. Units 2 and 3.
- Why not E? It pays the least of any row, but that comparison is not valid: its rate is quoted in different units. Next slide.

Only A, B, and C promise a schedule that is actually certain.

# Breakeven Inflation

B and E both come from the U.S. government and span ten years. B promises a fixed number of dollars, called nominal dollars. E's payments rise with inflation, so they are measured in real dollars. The difference between their rates is:

$$\underbrace{4.71\%}_{\text{nominal}} - \underbrace{2.41\%}_{\text{real}} = \underbrace{2.30\%}_{\text{breakeven inflation}}$$

That is the ten-year breakeven inflation rate: the inflation rate at which B and E pay you the same amount. The Federal Reserve publishes this rate directly, and its value for the same day is 2.30%. It is not a pure forecast, because it also reflects how much investors must be paid to accept uncertainty about inflation.

Market prices contain forecasts. Learning to extract them is a major goal of this course.

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**Source:** FRED series `DGS10`, `DFII10`, `T10YIE`, as of 2026-08-18

# A Bank Failure in March 2023

In March 2023 a bank with about $200 billion in assets failed in two days. It held one of the five promises on your list.

## Which one, and why?

None of the five defaulted. Every payment that came due was made as promised, and the bank failed anyway.

The clip identifies which one. It does not explain why in a way you can check; week 2 supplies that explanation.

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**Clip:** PBS NewsHour, 28 March 2023, 6 minutes

# Why You Cannot Rank the Five Yet

Every promise on that list comes from the United States government or from large companies with long records of paying what they owe. No vote in this room was careless.

You cannot rank those five yet, and the reason is precise:

> You have no rule for comparing dollars that arrive on different dates.

$264 in thirty years and $10,028 in four weeks cannot be compared until we state what a dollar on one date is worth on another. We build that rule on Thursday, and it is the first thing this course does.

# Summary

This lecture placed the course in context, showed that investment exposure is widespread, and read five financial promises as lists of payments and dates.

## Key takeaways

- **Investment choices are widespread.** A retirement account requires someone to decide how the money is invested, even if its default mix remains unchanged.
- **Each choice is a payment schedule.** A bill, a note, a bond, a corporate bond, and an inflation-indexed security are labels for lists of payments and dates, though not every list is equally certain.
- **Testing happens in stages.** Theory, backtest, forward test, and real money use different evidence, and each can reveal a different kind of failure.

On Thursday we build the rule that lets us compare those five payment schedules.

# Further Reading

Not assigned, and not on any problem set. These four sources describe the Federal Reserve System, the committee mentioned earlier, and the Treasury.

- *The Structure and Functions of the Federal Reserve System*, Federal Reserve Board: the roles of the Board of Governors, the Reserve Banks, and the FOMC.
- *Economy at a Glance: Policy Rate*, Federal Reserve Board: the target range, the effective rate, and the current implementation tools.
- *Fedpoints: Federal Reserve System*, Federal Reserve Bank of New York: an institutional overview.
- *Role of the Treasury*, U.S. Department of the Treasury: fiscal, financial, and debt-management responsibilities. The Treasury is a different institution from the Federal Reserve; week 2 distinguishes them.